

Executive Summary

This project focuses on the classification of iris flower species using Machine Learning techniques. The main objective of the project is to predict the species of an iris flower based on four important features: sepal length, sepal width, petal length, and petal width. The dataset used for this project is the well-known Iris Classification Dataset obtained from Kaggle.

The project began with Exploratory Data Analysis (EDA) to understand the structure and distribution of the dataset. Different visualizations such as pairplots and heatmaps were created to analyze feature relationships and class separability among the flower species. The dataset was checked for missing values and data consistency before model training.

Three classification algorithms were implemented and compared:

- K-Nearest Neighbors (KNN)
- Logistic Regression
- Decision Tree Classifier

The dataset was divided into training and testing sets using train-test split methodology. Each model was trained using the training data and evaluated on the testing data. Performance metrics including accuracy score, confusion matrix, precision, recall, and F1-score were used to evaluate model effectiveness.

Among all algorithms, the K-Nearest Neighbors (KNN) model achieved the best performance and accuracy for iris species prediction. The trained model was saved using the Joblib library in `.pkl` format to allow future predictions without retraining the model.

This project demonstrates the complete machine learning workflow including data preprocessing, visualization, model training, evaluation, and deployment preparation. It also highlights the importance of comparing multiple algorithms to identify the best-performing model for a classification problem.

The final deliverables of the project include the Jupyter Notebook, saved machine learning model, screenshots of outputs and visualizations, README documentation, and GitHub repository link.